

ARNAV CHUDIWALE

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PROFESSIONAL SUMMARY

MS Industrial Engineering student combining optimization modeling, supply chain analytics, and operational systems thinking to support data-driven decision-making. Skilled in translating business constraints into mathematical models using Gurobi, OR-Tools, and network optimization with proven applications in reverse logistics, last-mile delivery, and AI feasibility analysis. Pursuing operations and supply chain analytics roles in operations-intensive organizations.

EDUCATION

Master of Science, Industrial Engineering & Management Stillwater, OK
Oklahoma State University Jan 2025 – Dec 2026 (*Expected*)

Relevant Coursework: Operations Research, Network Optimization, Stochastic Processes, Discrete Event Simulation Methods, Supply Chain Strategy, Advanced Data Analytics for Industrial Engineering, Advanced Production & Inventory Control, Fundamentals of GIS

Bachelor of Engineering, Electrical & Computer Engineering Patiala, India
Thapar Institute of Technology Nov 2020 – Nov 2024

TECHNICAL SKILLS

Programming & Analytics: Python, SQL, R, Excel (Advanced: Solver, VBA, Power Query)

Optimization & Modeling: Gurobi Optimizer, Google OR tools, PuLP

Simulation & Supply Chain: Simio, Supply Chain Guru, Monte Carlo simulation

Data Visualization & GIS: Power BI, Tableau, Python (matplotlib, seaborn, plotly), ArcGIS Pro (Network Analyst, spatial analysis)

Cloud & Enterprise: Azure Cloud, Azure OpenAI

EXPERIENCE

AI Operations & Feasibility Analysis Mumbai, India
Ernst & Young (EY) LLP – Government of India, Department of Agriculture Jan 2024 – Jun 2024

- Supported deployment planning for statewide GenAI-based crop advisory platform serving 2M+ farmers by modelling system load scenarios and operational cost projections across 5 adoption rate scenarios (20%-80% penetration)
- Conducted cost-benefit analysis using Azure cloud pricing and usage forecasts to estimate monthly operating costs of (INR) 8.5M - (INR) 42M, enabling Department of Agriculture to approve phased 3-year roll-out with 30% annual budget increase
- Built financial model projecting 3-year Total-Cost-Of-Ownership (TCO) under different adoption curves; recommended staged deployment approach prioritizing high-yield districts to manage infrastructure risk and optimize ROI during scale-up phase

PROJECTS

Last-Mile Delivery Optimization for Peak Season Surge Capacity Dec 2025

- Developed a multi-strategy optimization model for last-mile delivery operations handling 162% peak season demand surge (1,490 → 3,906 packages/day), achieving 100% service level and reducing cost-per-package by 22% through hybrid overtime-outsourcing allocation using Google OR-Tools VRP solver
- Designed a dynamic routing simulation with time-based vehicle dispatch that reduced peak-day operational costs by 40.8% (\$10,780/day savings) and cut 3PL outsourcing by 54% (1,922 → 892 packages) by enabling fleet reuse across multiple dispatch waves without additional capital investment
- Built end-to-end supply chain analytics pipeline processing 500 urban delivery locations with sensitivity analysis across demand variance ($\pm 20\%$), 3PL capacity constraints, and overtime feasibility—delivering actionable trade-off insights that avoided \$41,508/day in late delivery penalties

Circular Supply Chain Network Design for Electronics Reverse Logistics & Refurbishment Jan 2026

- Engineered an optimized circular supply chain for a California electronics retailer using Gurobi (MIP) and ArcGIS Network Analyst, transforming ~28,000 annual smartphone returns from a disposal cost into a profit center with a \$7.95M 3-year NPV and a 7-month payback period
- Developed a multi-criteria facility location model, reducing total system costs to \$1.55M/year while achieving a 541% increase in value recovery through grade-based refurbishment
- Designed an end-to-end data pipeline in Python and integrated results into Power BI dashboards to visualize sensitivity analysis, proving project viability despite $\pm 20\%$ fluctuations in resale prices and return volumes